

TH-AMP01: Single-Supply Discrete Operational Amplifier

Electrical and Electronics Engineering Student Project

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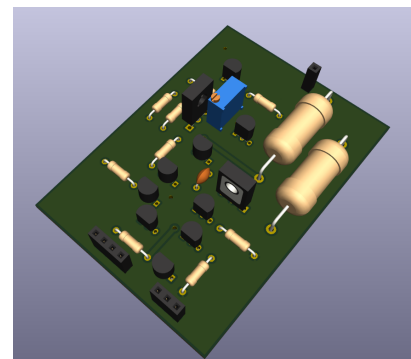


Figure 1: TH-AMP01 3D Render

1 General Description

The TH-AMP01 is a single-supply, discrete operational amplifier module designed for educational and general-purpose analog signal processing. Built entirely from standard through-hole BJT components, it features a differential input stage with an active current mirror load, followed by a Class-AB push-pull output stage. It is optimized to operate on a standard 9 V DC supply but can accommodate a wide range of voltages from 5 V up to 15 V (up to 20 V with thermal management and bias calibration).

2 Device Body Information

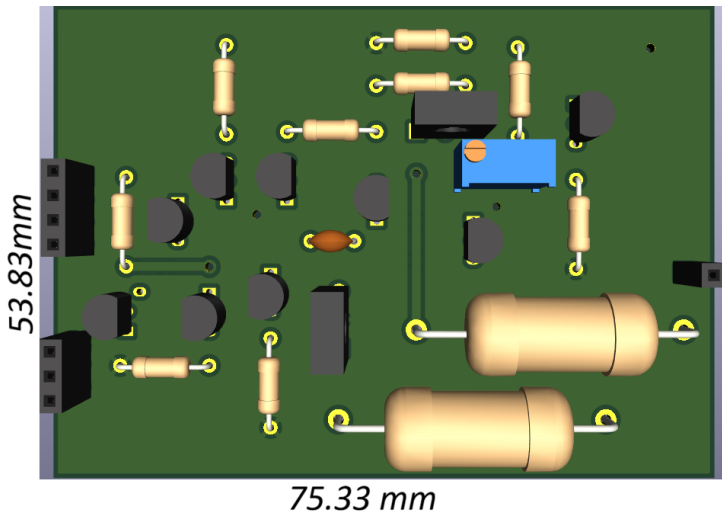


Figure 2: TH-AMP01 Top View (X -axis: 75.33 mm, Y -axis: 53.83 mm)

The TH-AMP01 is built on a custom compact printed circuit board (PCB) with physical dimensions of 75.33 mm width (X -axis) and 53.83 mm height (Y -axis). It utilizes a standard 2.54 mm pitch pin-header interface layout, making it fully compatible with standard breadboards and project enclosures.

3 Features

- **Fully Discrete Architecture**
- **Wide Supply Voltage Range:** from 5 V up to 15 V (up to 20 V with thermal management and bias calibration).
- **Class-AB Output Stage**
- **Open-Loop Topology**

4 Applications

- Academic analog electronics projects and laboratories

- Custom guitar practice amplifiers (driven by external Buffer, EQ, and Gain stages)
- Audio pre-amplification and basic active filtering
- Radio output stage

5 Schematic and Pin Configuration

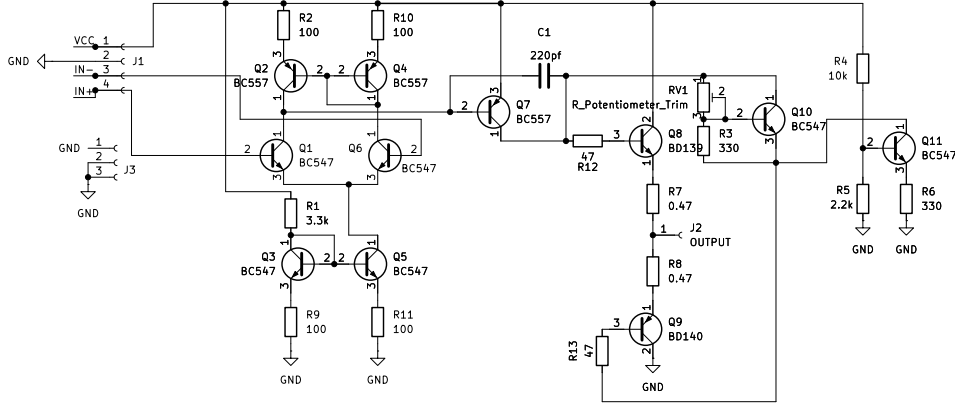


Figure 3: TH-AMP01 Schematic Diagram

Pin Name	Type	Description
IN+	Input	Non-inverting input
IN-	Input	Inverting input
VCC	Power	Positive power supply voltage
GND	Power	Ground reference
OUT	Output	Amplifier output signal

6 Absolute Maximum Ratings

Parameter	Symbol	Maximum Value	Unit
Supply Voltage	V_{CC}	20 ⁽²⁾	V
(2) Continuous operation at 20 V requires recalibration of the bias network via the onboard trimpot and adequate aluminum heatsinks on the output transistors. Input Voltage	V_{IN}	-0.3 to V_{CC}	V
Output Short-Circuit Duration	t_{SC}	Continuous ⁽¹⁾	—
Operating Free-Air Temperature	T_A	0 to 70	°C
Storage Temperature	T_{stg}	-40 to 125	°C

(1) Not bench-verified; output stage protection relies solely on emitter ballast resistors R6/R11, there is no active current-limit circuit. Treat as a design goal, not a guaranteed rating, until tested.

7 Recommended Operating Conditions

Parameter	Min	Typ	Max	Unit
Supply Voltage (V_{CC})	5.0	12.0	15.0	V
Load Impedance (R_L)	8	–	–	Ω
Operating Free-Air Temperature (T_A)	0	25	50	$^{\circ}\text{C}$

8 Electrical Characteristics

Measured at $T_A = 25^{\circ}\text{C}$, $V_{CC} = 9\text{ V}$ (unless otherwise noted).

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
I_Q	Quiescent Current	$V_{IN} = 0\text{ V}$, No Load	11		13	mA
P_{OUT}	Output Power	$V_{CC} = 9\text{ V}$, $f = 1\text{ kHz}$, $R_L = 8\Omega$,	–	336	–	mW
$V_{O(max)}$	Maximum Output Swing	$f = 1\text{ kHz}$, $A_V = 11$, $R_L = \text{Open}$	–	7.26	–	V_{pp}
SR	Slew Rate	$A_V = 11$, 10 kHz square-wave	–	4.98	–	$\text{V}/\mu\text{s}$
BW	Closed-Loop Bandwidth	$A_V = 11$, -3 dB point	–	620	–	kHz
GBW	Gain-Bandwidth Product	$A_V = 1$	–	6.82	–	MHz
THD	Total Harmonic Distortion	$f = 1\text{ kHz}$, $A_V = 11$, No Load	–	0.1	–	%
$PSRR$	Power Supply Rejection Ratio	$f = 1\text{ kHz}$, Simulated	–	51	–	dB
I_{BIAS}	Input Bias Current	$V_{IN} = 0\text{ V}$	–	300	–	nA

9 Typical Characteristics / Performance Curves

Vout (dB) ve Frequency (Hz)

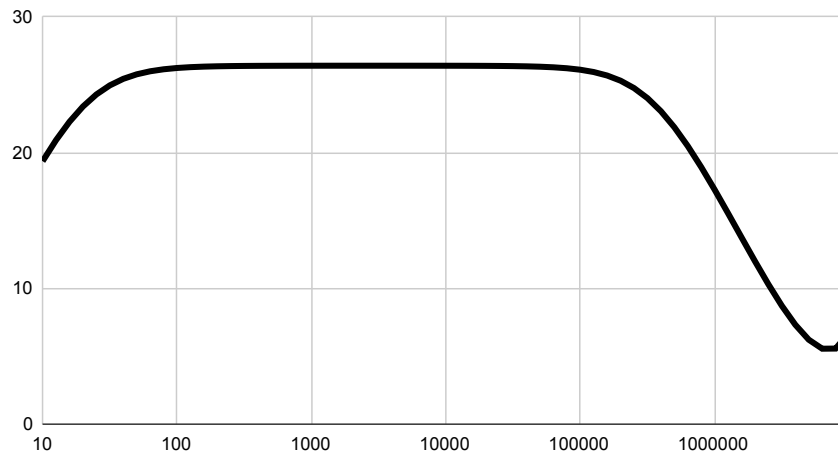


Figure 4: Frequency Response (Bode Plot) – from simulation

Supply Current(mA)-Supply Voltage (V)

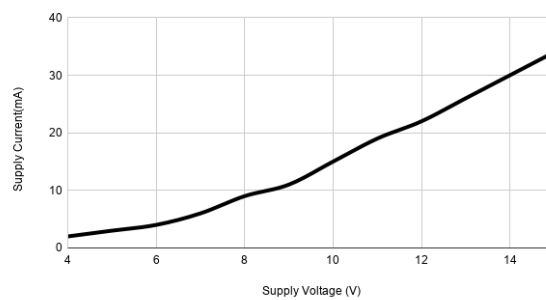


Figure 5: Supply Current vs Supply Voltage

Supply Voltage (V)-Max Output (Vpp)

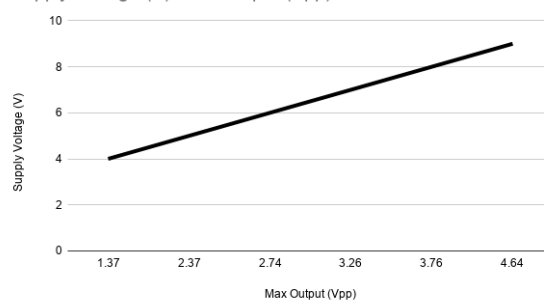


Figure 6: Maximum Output Swing vs Supply Voltage.

10 Detailed Description

10.1 Overview

The TH-AMP01 uses a three-stage discrete op-amp topology on a single 5–24 V supply: (1) a differential input pair with active current-mirror load, (2) a common-emitter voltage-amplification stage (VAS) with Miller compensation, and (3) a complementary Class-AB emitter-follower output stage.

10.2 Functional Block Diagram

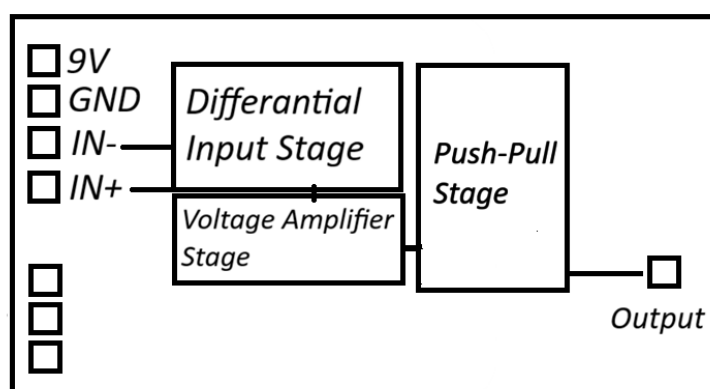


Figure 7: TH-AMP01 Functional Block Diagram

11 Application and Implementation

Note: Information in the following section reflects student-project-level simulation and design work; it has not been independently verified or warranted. Users should test and validate their own implementation.

11.1 Application Information

The TH-AMP01 is intended to be wrapped with external front-end circuitry (buffer / EQ / gain) and a feedback network, then used as the power output stage of the overall system. Because the device is an **open-loop, single-supply** amplifier, four things must always be added externally before it can be used, regardless of the specific application. Figure 11.1 shows the minimum required external network; Section 11.2 onward gives worked, component-value examples built on this template.

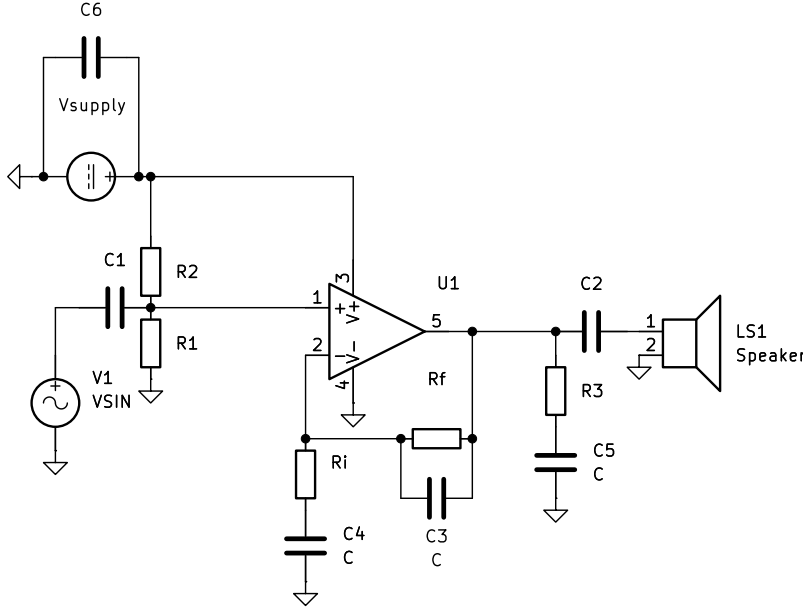


Figure 8: General external component requirements for TH-AMP01

- **Input Bias Network and Input Impedance (R_1 , R_2):** Forms a voltage divider to bias the non-inverting input to $V_{CC}/2$. This is essential for single-supply operation to ensure a symmetrical output voltage swing. Also these resistors adjust input impedance. Input impedance = $R_1 // R_2$.
- **Feedback Network (R_f , R_i):** Sets the closed-loop voltage gain of the amplifier. The AC voltage gain is approximately calculated as $A_V \approx 1 + R_f/R_i$.
- **DC Blocking Capacitors (C_4 , C_1):** C_1 blocks external DC from the input source, while C_4 reduces the DC gain to unity to preserve the $V_{CC}/2$ bias point. Both should be large enough (e.g., $100\ \mu\text{F}$) to pass low audio frequencies.
- **Compensation Capacitor (C_4):** Placed in parallel with R_f to improve phase margin and prevent high-frequency oscillations. A small value like $5\ \text{pF}$ is recommended.
- **Output Coupling Capacitor (C_2):** Blocks the internal $V_{CC}/2$ DC offset from reaching the load. If driving a low-impedance load like an $8\ \Omega$ speaker, a very large value such as $1000\ \mu\text{F}$ must be selected.
- **Feedback Compensation Capacitor (C_3):** Placed in parallel with the feedback resistor R_f . It provides phase lead compensation, improving the phase margin and preventing high-frequency oscillations. A small value like $5\ \text{pF}$ to $22\ \text{pF}$ is recommended.
- **Supply Bypass Capacitor (C_6):** Connected directly across the power supply (V_2). It acts as a decoupling capacitor, providing a low-impedance path to ground for high-frequency noise from the power supply. It also serves as a local energy

reservoir, allowing the amplifier to handle fast transient current demands smoothly.

11.2 TH-AMP01 with Gain = 21

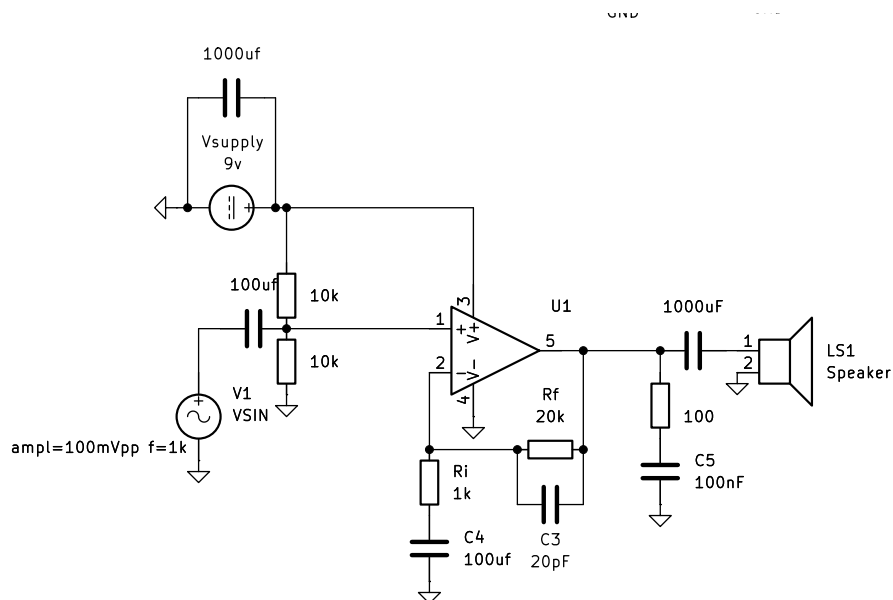


Figure 9: gain-of-21 audio amplifier configuration

11.2.1 Design Requirements

Design Parameter	Example Value
Load Impedance	$8\ \Omega$
Supply Voltage	9 V
Target Closed-Loop Gain	21 ($\approx 26.4\ \text{dB}$)

11.2.2 Application Curve

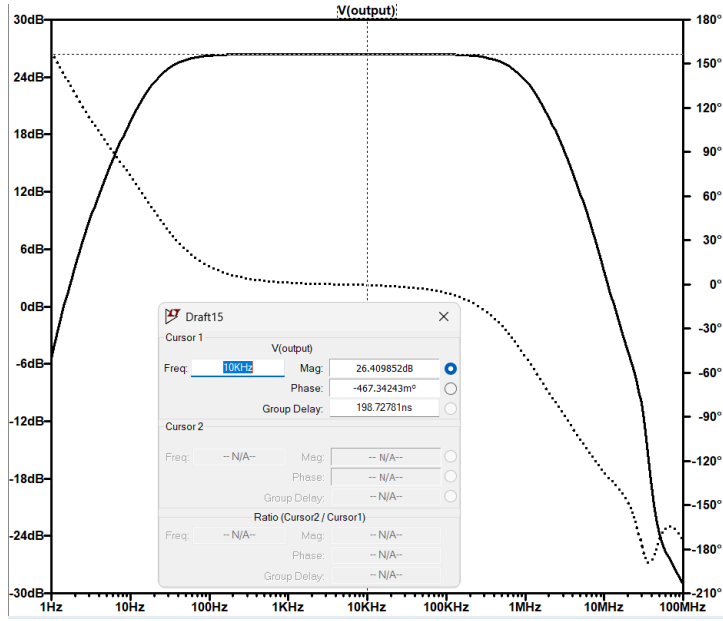


Figure 10: Closed-Loop Voltage Gain vs Frequency, Gain = 21 configuration

12 Power Supply Recommendations

- **Use a Clean Power Supply:** The module does not have heavy internal filtering. Any noise, hum, or ripple from your power supply will directly leak into the amplifier's output.
- **Add Decoupling Capacitors:** Always place a large electrolytic capacitor (e.g., 100 μ F) and a small ceramic capacitor (e.g., 100 nF) in parallel, as close to the VCC and GND pins as possible. This prevents sudden power drops and stops the amplifier from producing unwanted oscillations.

13 Layout

13.1 Layout Guidelines

To ensure optimal performance and stability, the PCB design follows these key principles:

- **Symmetric Differential Pair:** The input transistors (Q3, Q4) are routed with matched trace lengths to minimize input offset errors.
- **High-Current Traces:** Short and wide traces are used for the output stage and OUT pin to handle audio power levels without significant voltage drops.
- **Solid Grounding:** A continuous copper ground pour is utilized to keep the high-current return paths separate from the sensitive input signals.

13.2 Layout Example

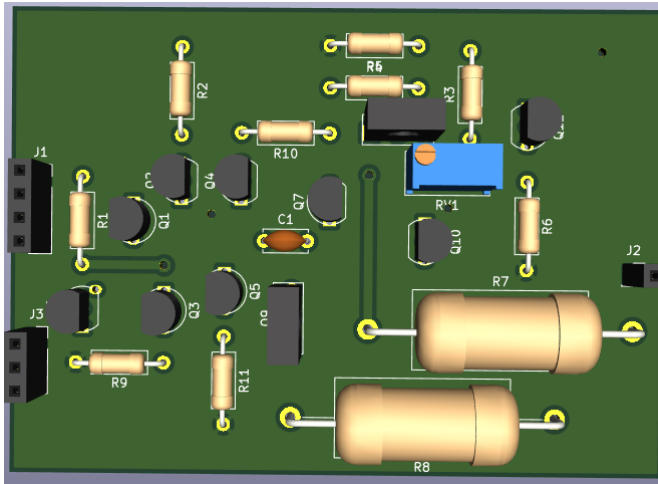


Figure 11: TH-AMP01 PCB Top Layer – Component Placement

14 Device and Documentation Support

This is a student educational project and is not a commercially supported product. No warranty, express or implied, is provided. All simulation results are based on generic SPICE models and should be treated as approximations of real device behavior.